MIDIKnob XL
Assembly Manual 1.0

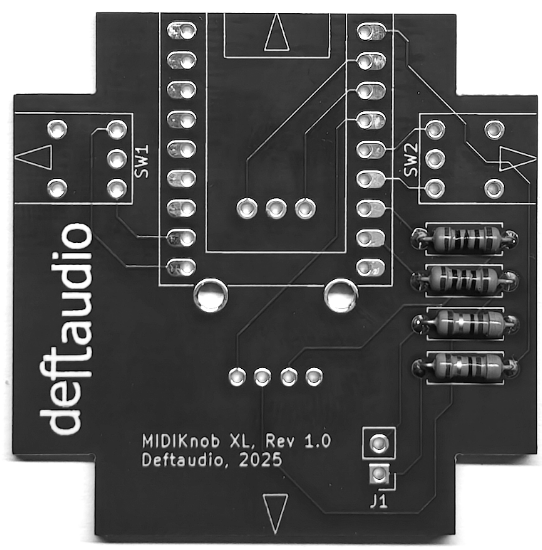
Overview

The MIDIKnob XL is a compact, yet powerful, programmable encoder controller designed for both MIDI control and everyday system interaction.
This is a **medium-difficulty kit** that requires soldering and firmware programming. Optional enclosure painting is also possible. The average assembly time is **25–30 minutes**.

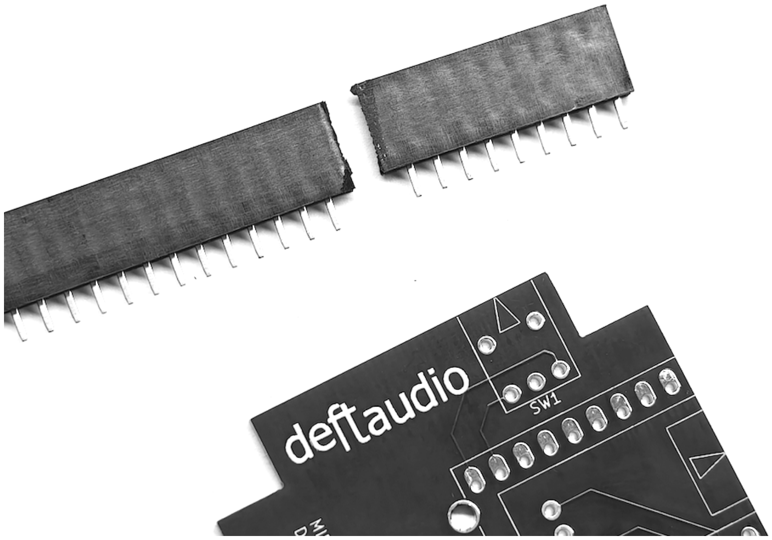
Hardware Assembly

Start by inspecting your package and verifying the received parts against the Bill of Materials (BOM) list.
Follow these steps for hardware assembly:

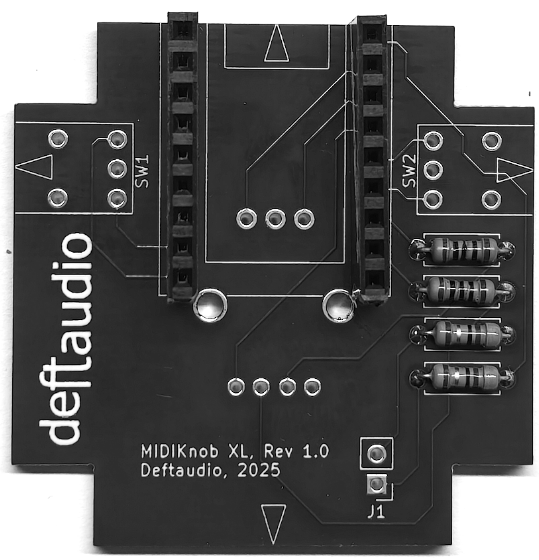
1. **Populate Resistors:** Solder the two value resistors (100 Ohm and 47 Ohm) onto the Printed Circuit Board (PCB).



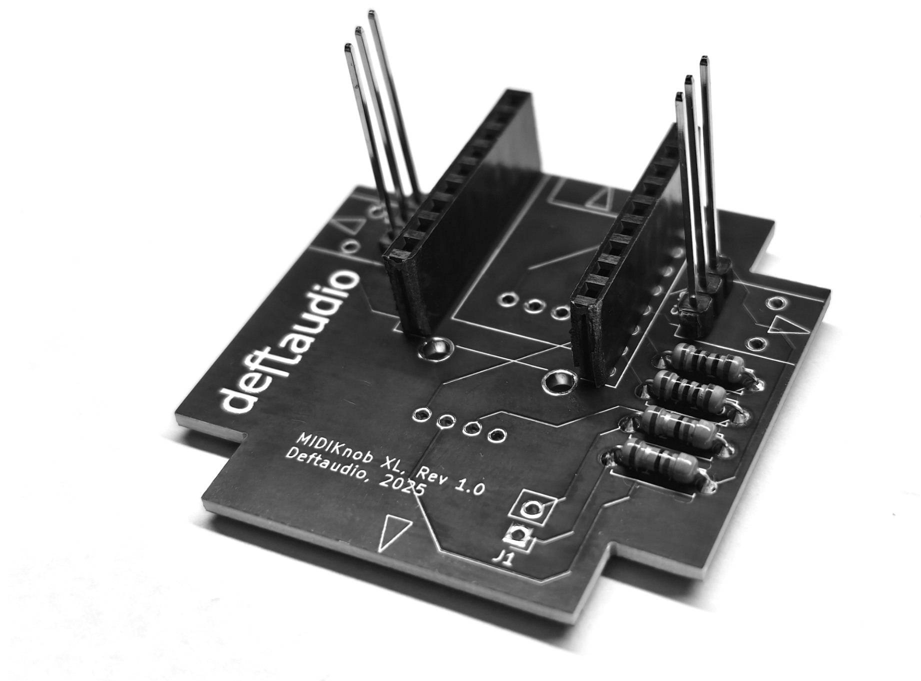
2. **Prepare Female Pin Strips:** Separate a female 40-pin strip into two pieces of 9 pins each.



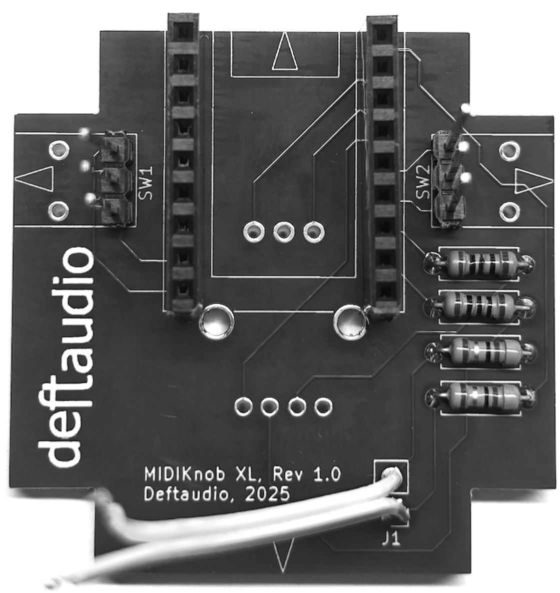
3. **Solder Female Pin Strips:** Solder the two 9-pin strips to the board.



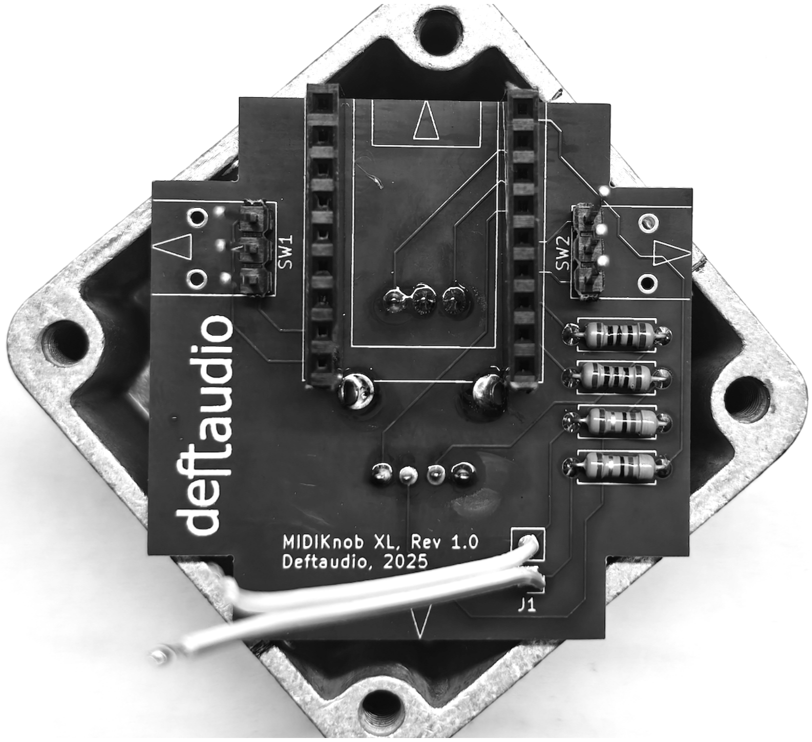
4. **Solder Side Headers:** Solder the 3-pin long headers into the switch positions on the sides of the board.



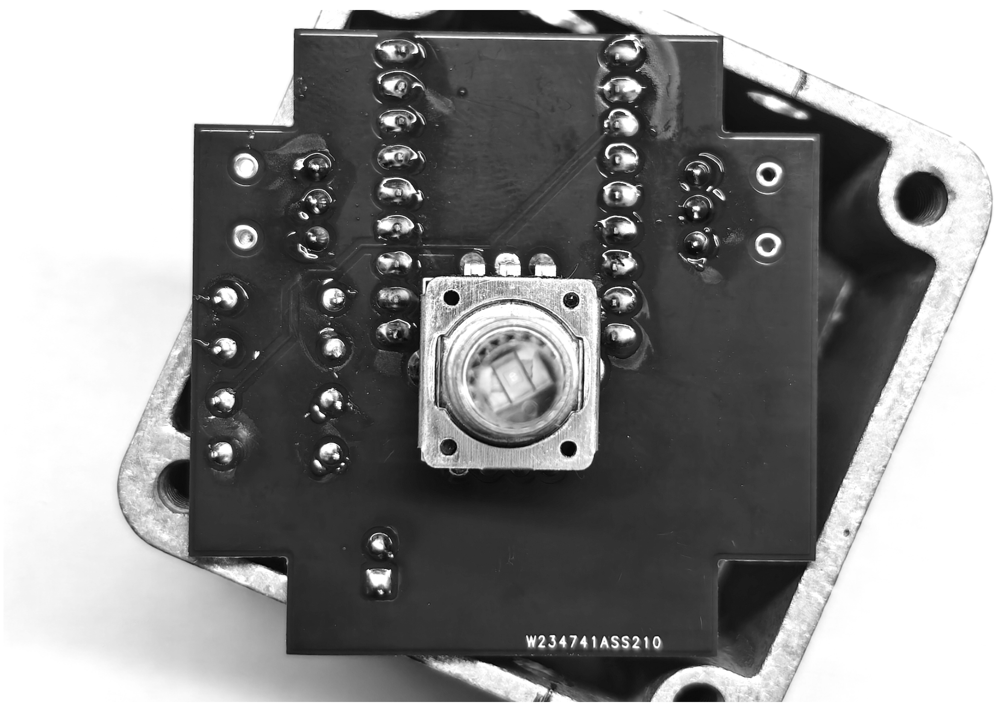
5. **Solder J1 Wires:** Solder two wires to the J1 pads.



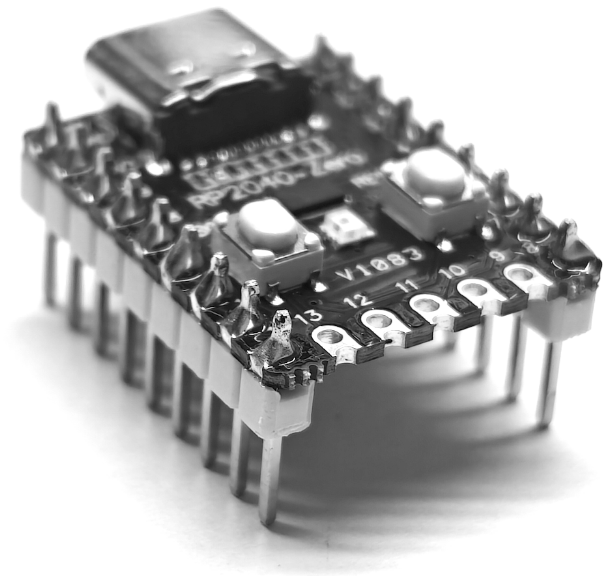
6. **Solder Encoder:** Solder the encoder from the bottom side of the board.



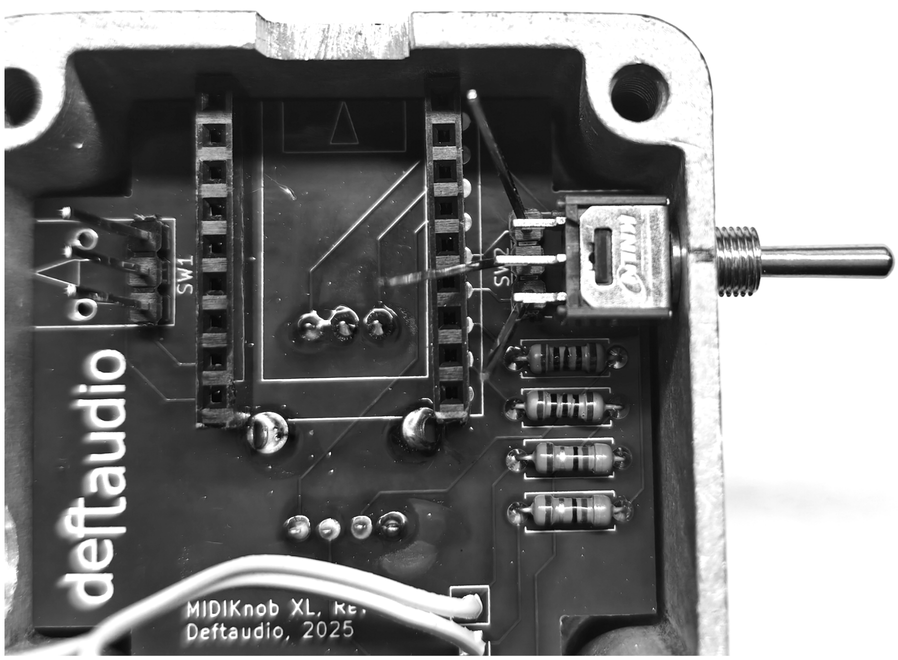
7. **Check Encoder Clearance:** Ensure the encoder case does not short any pins on the bottom side.



8. **Solder RP2040 Headers:** Solder the 9-pin male headers to the RP2040 board.

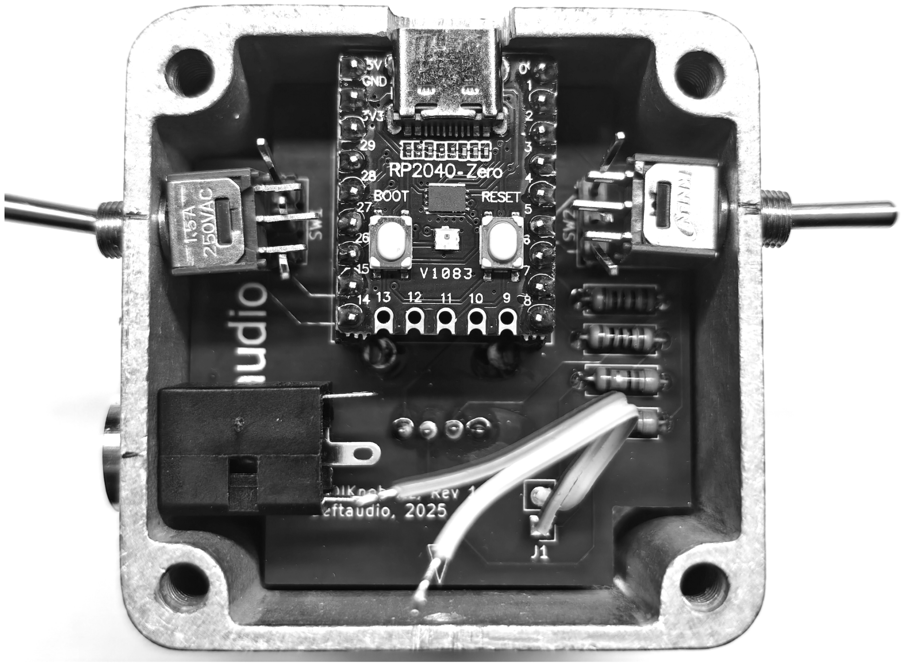


9. **Install Switches:** Insert the PCB into the enclosure. Spread out the switch pins and insert the first switch. This is easier to do before the RP2040 is inserted, as it provides more working room.



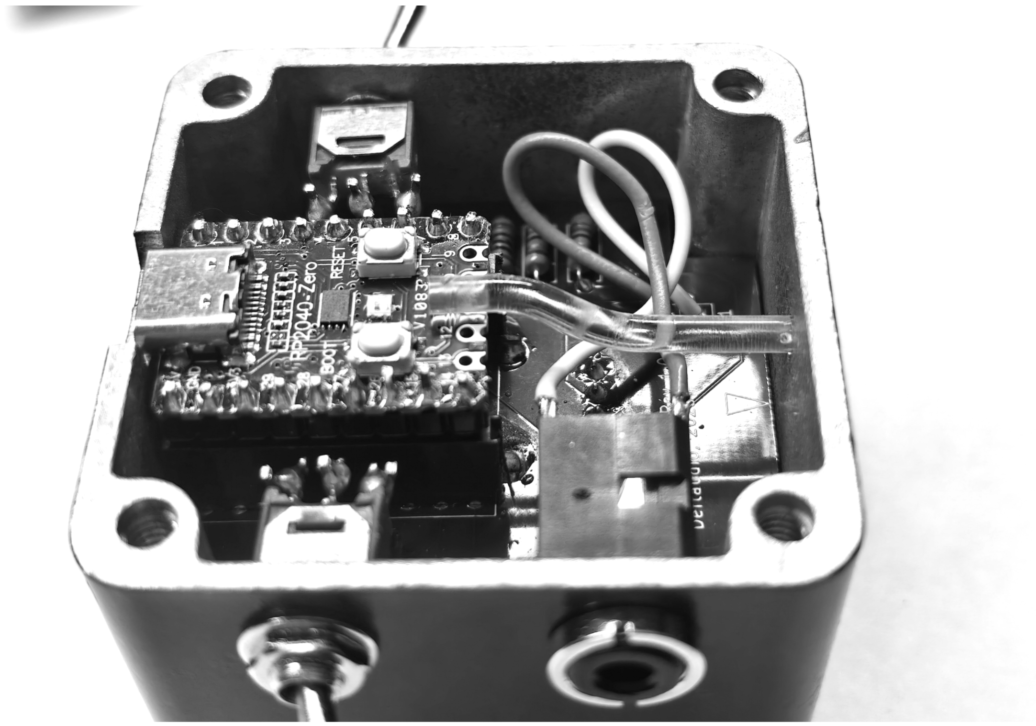
10. **Secure and Solder Switches:** Once installed, tighten the nuts on both switches and straighten the pins. Solder the pins to the switch legs, then carefully cut the excess pins so the lid can close properly.

11. **Install MIDI Output:** Install the MIDI Output TRS (Tip/Ring/Sleeve) socket and tighten its nut.

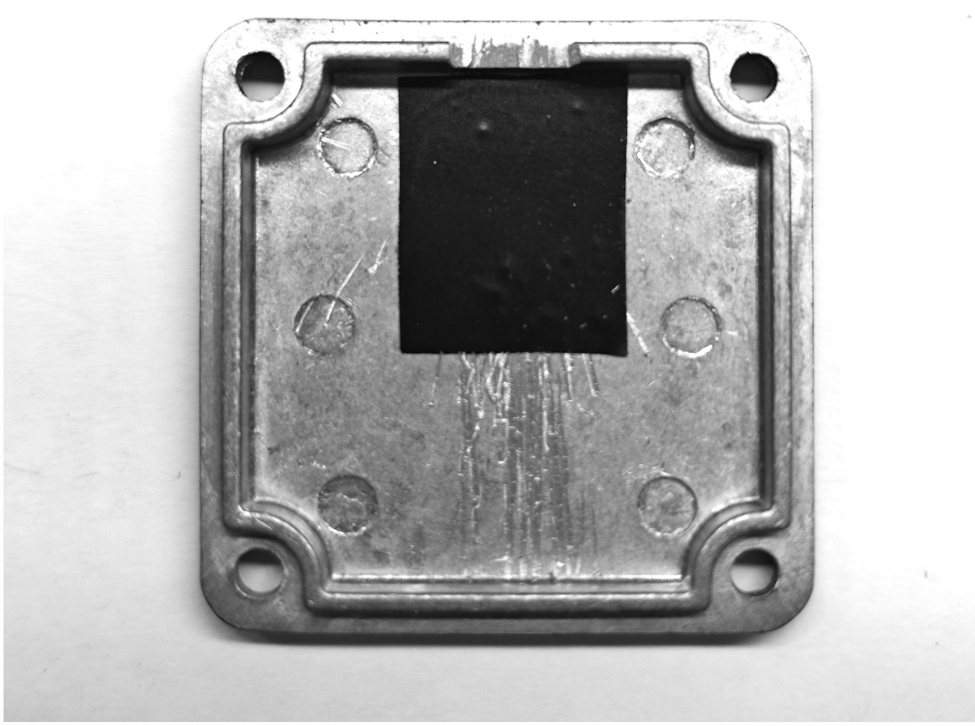


12. **Solder MIDI Wires:** You only need to solder two wires to the tip and ring pads (leave the sleeve unconnected). There are two MIDI TRS specifications: TRS-A (Source on tip) and TRS-B (Sink on tip). Solder according to your preference. *Hint: The J1 square pin is Source, the round pin is Sink.*

13. **Install Light Guide:** Finally, insert the clear plastic light guide. Heat it up with a lighter or a heat gun and bend it to the shape shown below.



14. **Isolate Bottom Lid:** Ensure the bottom lid does not short any pins. Place a piece of isolation tape to protect the pins from accidental shorting.



Firmware Programming

The MIDIKnob family runs on **CircuitPython**, a real-time Python environment specifically designed for the Raspberry Pi Pico (RP2040). CircuitPython is ideal for rapid prototyping and is easy to learn. The MIDIKnob code is open-source Python under the BSD-3 license, and contributions are highly encouraged.
To deploy the firmware, you must first install CircuitPython on a blank RP2040 and then copy over the Python code files.

Installation Steps:

- **Download CircuitPython:** Download the latest CircuitPython UF2 file for the RP2040_zero board from https://circuitpython.org/board/waveshare_rp2040_zero/.
- **Connect Device:** Connect the MIDIKnob to your PC or Mac. It will appear as a USB flash drive.
- **Install UF2:** Copy the downloaded UF2 file onto the USB flash drive. The device will automatically self-reset and reappear as a new USB flash drive containing system files.
- **Download Firmware:** Download the MIDIKnob Firmware and unpack the zip archive from https://github.com/Deftaudio/Midi-boards/tree/master/MidiKnob/FW.
- **Copy Firmware Files:** Replace the content of the Pico USB drive with the content of the unpacked archive.
- **Finalize:** Safely disconnect and reconnect the MIDIKnob.

Once reconnected, the MIDIKnob code will be active and running. You will see the new startup sequence, which triggers various LED colors. However, the device will lack configuration in its non-volatile memory, so basic MIDI programming is needed.

Operating Modes:
The device supports three operating modes, selectable via the switch on the right side:

- **MIDI Mode**
- **System Volume**
- **Keyboard Shortcut Mode**

In each mode, the switch on the left allows you to select one of **three stored presets**, providing instant access to multiple configurations.

MIDI Mode

In MIDI Mode, rotating the encoder sends a **MIDI CC (Control Change) message** on the assigned controller number. Pressing the encoder sends a **Note On / Note Off** message.
The device also responds to incoming **MIDI Clock**, causing the encoder's LED to pulse on the start of each bar and on every beat, providing clear visual timing feedback.

MIDI Programming: Programming MIDI behavior is quick and intuitive: simply **double-tap the encoder** and send a MIDI CC or Note On message from an external controller. The assignment is automatically saved to **non-volatile memory** and recalled upon power-up.

System Volume / Keyboard Mode

In System Volume mode, the device operates as a **USB HID (Human Interface Device) controller**, providing native control of your computer's audio volume. Rotating the encoder adjusts the volume up or down. Pressing the encoder sends a default keyboard command, which can be remapped as needed.

Customizing Firmware

To make changes to the device's behavior, you can edit the Python code:

1. **Open `Code.py`:** Use an external editor to open the `Code.py` file on the MIDIKnob USB drive.
    - **Recommendation:** Thonny is a simplistic editor that allows for easy changing, restarting, and configuring of the Pico board.
2. **Make Basic Changes:** Simple modifications, such as changing the keyboard shortcuts the MIDIKnob sends, involve finding the relevant line in the code and modifying it.

If you are not comfortable programming yourself, you can use an AI coding assistant such as Codex from ChatGPT. It's free for small projects. Start a new project, point the assistant to the MIDIKnob files, and ask it to make the changes for you. This approach works perfectly for straightforward code modifications and gives an immediate result.